

Guided Detection of Hidden Subclusters in a High-Dimensional Data Space via Interactive Visual Analytics

Filmon Mesgun

Filmon.Mesgun@studmail.htw-aalen.de

Aalen University of Applied Sciences
Aalen, Germany

Julia Winter

Julia.Winter@studmail.htw-aalen.de

Aalen University of Applied Sciences
Aalen, Germany

Andreas Theissler

Andreas.Theissler@hs-aalen.de

Aalen University of Applied Sciences
Aalen, Germany

ABSTRACT

With the continuous growth of data and its describing dimensions, data mining is essential to understand information and use it in a domain-specific context. Based on related attributes, potential groups can be established. This method is known as the unsupervised machine learning algorithm "clustering". Identifying the optimal clustering model is a crucial factor for further processing and data-driven decisions. However, in high-dimensional data, overlapping or noisy features lead to hidden clusters in the original dataset. Because experts in their corresponding domain possess a deeper understanding of the data than regular machine learning algorithms, newest approaches consist of interactive visualisation solutions where experts can select relevant features and reduce the dimensionality of datasets. This paper presents a solution based on a bottom up search concept for subspace clustering. Further the tool supports the domain expert to identify the optimal clustering and to explore the data's composition and potential hidden clusters.

We combined Visual Analytics, subspace clustering and clustering methods to enable a knowledge-based feature selection for domain experts. Those insights are used as immediate feedback to the solution for enhancing the clustering results.

CCS CONCEPTS

• **Computing methodologies** → **Feature selection**; • **Clustering** → **High Dimensional Data**; • **Hidden Cluster**; • **Interactive Visual Analytics**;

KEYWORDS

Visual Analytics, Hidden Cluster, Subset, High Dimensional Data

ACM Reference Format:

Filmon Mesgun, Julia Winter, and Andreas Theissler. 2021. Guided Detection of Hidden Subclusters in a High-Dimensional Data Space via Interactive Visual Analytics. In *xxxxxxx '21: ConfX, January 26, 2021, xxxxxx, DE*. ACM, New York, NY, USA, 5 pages.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

xxxxxxx '21, January 26, 2021, xxxxxx, DE

© 2021 Association for Computing Machinery.

ACM ISBN 978-1-4503-XXXX-X/18/06...\$15.00

1 INTRODUCTION

Clustering is used to accumulate similar feature characteristics of data into distinctive groups. [10]. The literature differentiates between hard- and soft clustering [13]. The former associates data points to primarily one specific cluster, a standard method to apply hard clustering is "K-Means" [18]. Soft clustering is defined by membership values which evaluate the affiliation of each data point to existing clusters. This is conducted by "Fuzzy C-means Clustering" [3]. While these algorithms perform well for low-dimensional spaces, an increase of dimensions poses difficulties to detect clusters accurately [23]. The following conditions cause this: the distance between the data points converges, and after reaching a certain extent, these points are indistinguishable from each other [20]. Furthermore, subspaces may have different clusters so that they cannot align to higher levels [1]. And lastly, the cluster can be hidden due to features that have overlapping data points in the original data space or due to noise caused by distorted data points [4]. To solve the latter issue and detect hidden cluster in a high-dimensional space, subspace clustering can be performed with top-down, and bottom-up search methods [20]. Combined with data visualisation, these methods support the expert to identify non-significant features that have to be removed for an optimal clustering result [16]. However, to avoid important information lost, feature reduction must be carefully applied. By including experts in the selection process, a focused evaluation of features can be conducted. Insights of the data provide the user with an understanding of the correlation between individual features and enable hidden relation detection - also known as hidden clusters.

The main contributions of this paper are:

- A guiding approach for identifying hidden clusters in a high-dimensional space, based on subspace clustering with the CLIQUE algorithm.
- An exploration and presentation tool of distorted features to find potential hidden clusters
- Supported expert interaction by choosing hyperparameter, exploring data and providing feedback.
- An executable prototype: MesWin_XplorationHiddenCluster

2 RELATED WORK

INFUSE of Krause et al. [15] is an example for using Visual Analytics (VisAn) methods to reduce dimensions interactively through the feedback of an expert and the prediction of essential features by machine learning algorithms.

Assent et al. [2] provided an approach where subspace clusters are visualised to grant the user an overview of the subspace clustering and analyse specific clusters in more detail. Based on several parameter settings, the user can observe changes in the data and detect the best possible clustering result.

Lee et al. [17] build a VisAn approach to explore daily behaviour of patients to find signs of the Alzheimer's disease. However, the approach is limited to this use case. The visualisation is based on 3D RadVis and parallel coordinates and the interaction between them as well as the interaction between the visualisation and the expert.

Lakshmi et al. [16] proposed an algorithm based on redundancy in subspace clusters. It removes all clusters that have duplicates and identifies the interesting subspace through dependency measure, including the Apriori search algorithm. VisAn, graphical exploration of the data, and experts' feedback are not part of this approach.

In Thrun's et al. approach [25], high dimensional data is initially reduced to two dimensions through a reduction method chosen by the user. The clustering step is executed after a reduction on these remaining two dimensions.

ExplorableClusterExplorer by Fezer et al. [9] is an interactive and explorative feature selection tool that supports user to identify the optimal clustering of a given dataset. By defining the detected clusters as labels for a supervised setting, feature importance of a Random Forest and local interpretable model-agnostic explanations are combined to improve the feature subset selection and overall performance of the clustering algorithm. The experimental results were conducted on two low-dimensional datasets.

To the best of our knowledge, our approach is the first to provide a visual representation to detect hidden clusters by identifying and removing distorted features. Thereby, the original dataset is divided into subspace cluster. A visual representation of features are used to identify distorted features by the expert. Comparing the discussed work, we enable the user to explore the data space's composition, the correlation between dimensions and the subspace cluster. This information and the user's domain knowledge can then be used to distinguish relevant from non-relevant features. Furthermore, we provide a chart of the clusters based on selecting the required number of clusters and relevant features by the expert. The selection is enhanced by analysing several graphics as well as chosen metrics to determine the optimal hyperparameters for clustering the feature set. To evaluate the effects of feature extraction regarding hidden clusters, a final clustering is conducted by the expert.

3 VISUAL ANALYTICS SYSTEM FOR HIDDEN CLUSTER

In our application, we combine VisAn, subspace clustering and common clustering algorithms to support domain experts to find hidden clusters (see *DEFINITION 1*).

DEFINITION 1. (HIDDEN CLUSTER) In a dataset D with feature space $F = \{f_1, \dots, f_N\}$ of N features (variables, data points P), hidden clusters emerge due to overlapping features F_N or distorted data points P in $P = \{p_1, \dots, p_N\}$.

Based on this visual exploration, the user can provide feedback to the VisAn tool as feature deselection. With this feedback, it is

possible to detect and eliminate distorting factors and achieve the best possible clustering for the whole dataset.

Figure 1 shows an abstract visualisation of how our solution was developed. As the first initial preprocessing step, the expert must eliminate non-significant features before uploading the data into the VisAn tool. In this step, also character- and string based features have to be transformed to numeric values. The preprocessed data is then used to perform an initial **subspace clustering** algorithm. We chose the subspace clustering algorithm CLIQUE presented by Agrawal et al. [1]. This algorithm performs a density-based bottom-up search method to identify subspaces in a high dimensional data space. Recursive scanning of the dataset removed all dimensions with a density below a given threshold.

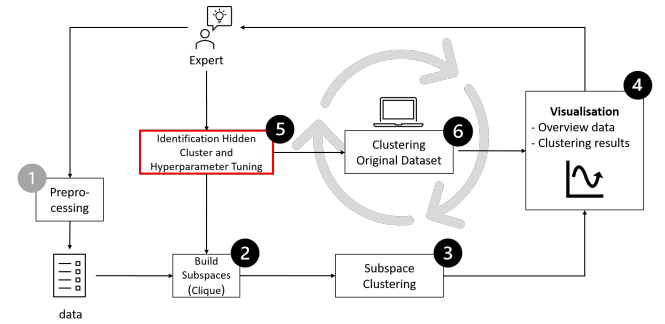


Figure 1: Our Model

1. Preprocessing of raw data by the expert, 2-3. Create subclusters by the CLIQUE algorithm, 4. Visualisation of the data and the subclustering or clustering results, 5. Experts identify distorted features to find hidden cluster based on the output and make parameter settings, 6. Clustering of most significant dimensions and interactive visualisation by cluster- and parallel plot, Steps 2-5/6 are in an iterative loop.

With the subsequent visualisation of step four, users can then tune the hyperparameters in step five, e.g. the threshold of CLIQUE and the number of clusters. Also, the clustering algorithm for step six can be selected. After every change, the model calculates again. Step two to six are processed in several loops until the best clustering is found and the user is confident with the clustering result. The application is implemented in R, version 4.0.2 [24] with shiny [5] and plotly [22]. Interactive visualisations were realised with the r2d3 package [19].

On the basis of the two-fold development of our tool, we provide two views of clustering results: the first page shows the subspace clustering results of the CLIQUE algorithm for the exploration and deselection of distorted features. The second page shows the results of the fuzzy or k-means algorithm performed on the previous restricted dataset for the observation and further optimisation.

3.1 Interactive Subspace Clustering

The tool loads an initial dataset, which can be changed if required. Parameters like features and density threshold for the CLIQUE algorithm can be set interactively. The feature list is used to reduce the dimensionality of the dataset. Feature deselection on the feature

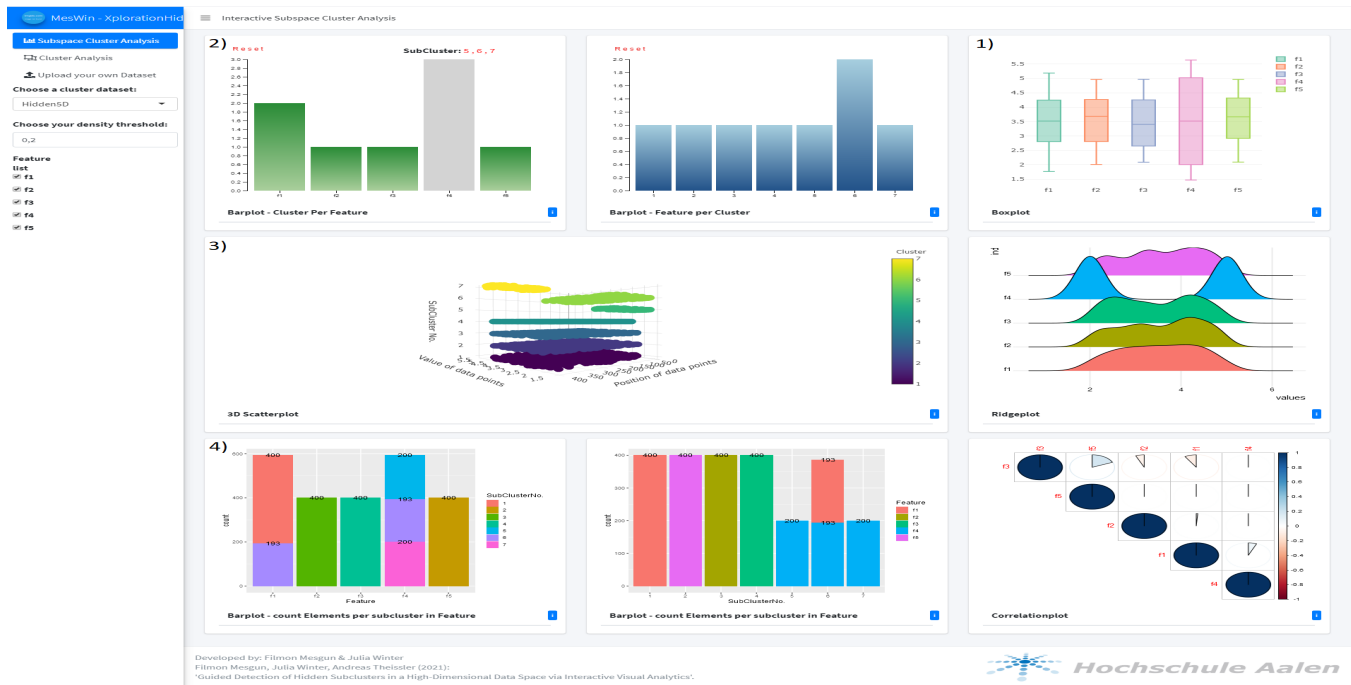


Figure 2: General Data Overview

Overview of the dataset and subclustering result. Left: navigation panel, selection of dataset, clustering algorithm, cluster amount and feature selection. Main panel: 1) the general data overview: boxplot, ridgeplot and correlation matrix, 2) subclustering result barplot cluster per feature and vice versa, 3) 3D scatterplot of selected features datapoints, 4) barplot of cluster amount per feature and feature amount per cluster.

list leads to an immediate change of the plots in this section and an in-depth analysis of specific dimensions. The CLIQUE algorithm partitioned a dataset into ξ equivalent blocks. These blocks are then analysed by their density. If a block exceeds or matches a certain percentage threshold, the block will be clustered [14, 20, 23].

Figure 2 provides an insight into the dashboard's first page. The visualisation is divided into three parts. The first part is the general data overview on the page's very right, provided as boxplot, ridgeplot and correlation plot. These plots provide a first orientation about the data. Outliers, conspicuousness in the distribution and coherent features can be recognised at first glance as well as identified as initial indicators for a potential distortion of features.

The second part represents the ratio between the subclusters calculated through the CLIQUE algorithm and the associated features.

Through this visualisation of our test dataset, we observed that when a features posses a high frequency of distorted datapoints, it is assigned to a higher quantity of subspaces in comparison to the other. On the basis of our test dataset, we discovered that features were well clustered if each feature belongs to an unique subcluster.

A 3D scatterplot was designed to show the subclusters compared to the position of the data points and their data point values in the original dataset. As exploration method, the 3D scatterplot is used to give insights about the subclusters and their composition of features and the distribution of the data points. With this, a discrepancy in the subspace clustering can be visualised and detected by the expert.

To avoid an overload of information in the 3D scatterplot we included two further barplots at the end of the page to enable an easy evaluation of the quantity of each feature in the subclusters and vice versa. The overview over the quantity of datapoints which were included from different subclusters in one feature and vice versa can be also an indicator of disturbing features.

Our dashboard solution also includes simplifications for the user to analyse the dataset. By clicking on one bar of the left barplot (Barplot - Cluster per Feature), the assigned subclusters are shown in an hover window and the scatterplot shows then only the data points of the selected feature. The general data overview changes to the features included in the selected cluster by clicking on one bar of the right barplot (Barplot - Feature by Cluster). With this step by step approach big datasets can be reduced to a size that benefits a good overall clustering result.

3.2 Overall Cluster Analysis

The results of the Interactive Subspace Clustering exploration page are then used for the subsequent part of the visualisation. A cluster plot of the remaining dimensions is calculated, as shown in Figure 3. In this part, the expert can choose clustering algorithms and the optimal number of clusters as additional interactive settings.

As potential clustering algorithms the "K-Means" [18] and "Fuzzy C-Means"-algorithm [3] are provided.

The algorithm results are displayed via a cluster plot and supported by several graphs to identify the clusters' optimal number.

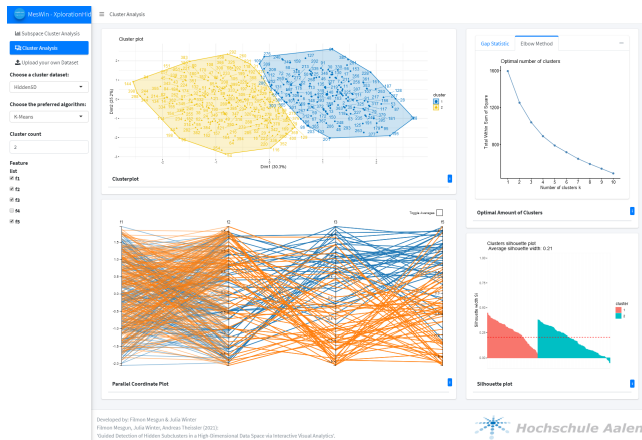


Figure 3: Overall Cluster Analysis Visualisation

Cluster plot for the visualisation of the clustering result. Gap Statistic and Elbow Method plot provide a calculated recommendation for the best cluster amount. Silhouette plot indicates the similarity between data points and their respective cluster. Parallel coordinate plot for further insights in the clustering results.

Furthermore, silhouette [21] and parallel coordinate plots are provided to measure the similarity and overall relationship of data points with each other and their respective clusters. As graphs for the optimal number of clusters, the Gap Statistic [26] and the Elbow Method [18] is used.

3.3 Evaluation

This section offers a walkthrough of the presented solution for detecting distorted features and the best overall clustering. To further evaluate our approach, we used two datasets. The first was an artificially generated dataset with synthetic generated noised features. Second, we assessed our approach with a heart failure prediction dataset that focused on the effects of serum creatine and ejection fractions [7].

The following walkthrough is based on the artificial generated dataset. As a preliminary step, the user uploads a dataset which needs to be clustered. The application begins with the first menu item "Subspace Clustering Analysis" to identify these. In Figure 2 the first barplot displays how many features a subcluster is assigned to. In our example, f4 is assigned to three distinctive subclusters: five, six and seven. Whereby the subclusters five and seven contain only feature f4. On the contrary, subcluster six consists of f1 and f4. Therefore, we identify f4 as the noisy feature that leads to the forming of hidden clusters. As an additional indicator, the features' distribution can be compared in the ridge plot on the right side. Visual differences can be observed in the cases of f1 and f4 in comparison to the other features. Due to the inconsistent subclustering of f4 the data points were split and clustered into different subclusters. This is displayed in the structure of the 3D scatterplot, where the expert can directly identify the noisy features. After evaluating each visualisation, feature f4 can finally be identified as the distorted feature that leads to the hidden data spaces and is required to be deselected in the feature list.

By excluding f4, the clustering algorithms can present exclusive subclusters and clusters for the dataset. The overall clustering is shown in the tab "Cluster Analysis", see Figure 3. Initially, the result of a k-means algorithm with a selected number of clusters of two is visualised. To evaluate the results and potentially detect better hyperparameters, the expert has to identify the optimal number of clusters by studying the Gap Statistic or Elbow Method plot. In our example, the optimal number is two. We conclude that through the deactivation of feature f4, the clustering algorithm k-means and the optimal cluster number of two, an optimal unique clustering can be established.

4 CONCLUSION AND FUTURE WORK

We designed and evaluated our tool based on an artificially created dataset where distorted features were known. In addition to that, visualisation could be tested and evaluated on an to the developer familiar dataset. After evaluating different artificial distorted and clean data, we tested our dashboard on other real datasets. We developed a VisAn dashboard where every numeric data can be imported and analysed regarding noisy or distorted features which leads to hidden clusters.

As future work and limitation has to be mentioned, that our approach was only proved on one real dataset. So further different real datasets have to be tested to verify the approach. Also, the verification of our tool with real datasets could only be reviewed by non-domain experts. In further work, this should be done by real domain experts instead of software developer. CLIQUE algorithm was computed with predefined parameters; there is a need to validate testing with further parameter to confirm our theory. The CLIQUE algorithm itself is limited by long computation times. It has to be proved if other subclustering algorithms can also be used to visualise noisy features to detect hidden clusters and if the computation time can be improved. To validate our results, we will test our approach with other subspace clustering algorithms. These will include CLIQUE's successor in the bottom-up search of the static grid, namely known as ENCLUS [6], and of the adaptive grid bottom-up search, the MAFLA algorithm [11]. The preprocessing step where data should be prepared as numerical values for the algorithm can also be integrated into the solution in a further step so that the expert is not required to do this manually. The clustering algorithm of page two only includes k-means and fuzzy C-means algorithms. To further evaluate the efficiency of our approach, we plan to test the results on other algorithms. As an example DBSCAN [8] could be applied for larger datasets with noisy dimensions as well as an alternative hierarchical method to cluster by using the density- and grid based algorithm, namely known as ICEAGE [12], both can potentially be added to our tool.

REFERENCES

- [1] Rakesh Agrawal, Johannes Gehrke, Dimitrios Gunopulos, and Prabhakar Raghavan. 1998. Automatic Subspace Clustering Mining. In *ACM SIGMOD International Conference on Management of Data*. 94–105. <https://doi.org/10.1145/276305.276314>
- [2] Ira Assent, Ralph Krieger, Emmanuel Mueller, and Thomas Seidl. 2007. VISA : Visual Subspace Clustering Analysis. *ACM SIGKDD Explorations Newsletter* 9, 2 (2007), 5–12. <https://doi.org/10.1145/1345448.1345451>
- [3] James C. Bezdek, Robert Ehrlich, and William Full. 1984. FCM: The Fuzzy c-Means Clustering Algorithm. *Computers & Geosciences Vol. 10*, 2-3 (1984), 191–203. <https://doi.org/10.1109/igarss.1988.569600>

- [4] C. Bouveyron, S. Girard, and C. Schmid. 2007. High-dimensional data clustering. *Computational Statistics and Data Analysis* 52, 1 (2007), 502–519. <https://doi.org/10.1016/j.csda.2007.02.009>
- [5] Winston Chang, Joe Chang, JJ Allaire, Yihui Xie, and Jonathan McPherson. 2020. shiny: Web Application Framework. <https://cran.r-project.org/package=shiny>
- [6] Chun-Hung Cheng, Ada Waichee Fu, and Yi Zhang. 1999. Entropy-based subspace clustering for mining numerical data. In *5th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 84–93. <https://doi.org/10.1145/312129.312199>
- [7] Davide Chicco and Giuseppe Jurman. 2020. Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making* 20, 1 (2020), 1–16. <https://doi.org/10.1186/s12911-020-1023-5>
- [8] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. 1996. A density-based algorithm for discovering clusters in large spatial databases with noise. In *Knowledge Discovery and Data Mining (Kdd)*. 226–231. <https://doi.org/10.1016/B978-044452701-1.00067-3>
- [9] Eric Fezer and Dominik Raab. 2020. XplainableClusterExplorer : A novel approach for Interactive Feature Selection for Clustering. In *13th International Symposium on Visual Information Communication and Interaction*. 1–5. <https://doi.org/doi/10.1145/3430036.3430066>
- [10] Guojun Gan, Jianhong Wu, and Zijiang Yang. 2006. A fuzzy subspace algorithm for clustering high dimensional data. In *Lecture Notes in Computer Science*. 271–278. https://doi.org/10.1007/11811305_30
- [11] Sanjay Goil, Harsha Nagesh, and Alok Choudhary. 1999. *MAFLA: Efficient and scalable subspace clustering for very large data sets*. Technical Report. 1–20 pages.
- [12] Diansheng Guo, Donna J. Peuquet, and Mark Gahegan. 2003. ICEAGE: Interactive Clustering and Exploration of Large and High-Dimensional Geodata. *GeoInformatica* 7, 3 (2003), 229–253. <https://doi.org/10.1023/A:1025101015202>
- [13] L. Himmelsprach and S. Conrad. 2019. Fuzzy Clustering of High Dimensional Data with Noise and Outliers. In *International Joint Conference on Computational Intelligence*. Springer Nature Switzerland AG, 221–262. https://doi.org/10.1007/978-3-540-28974-6_13
- [14] Karin Kailing and Hans-peter Kriegel. 2004. Density-Connected Subspace Clustering for High-Dimensional Data. In *4th SIAM International Conference on Data Mining*. 246–256. <https://doi.org/10.1137/1.9781611972740.23>
- [15] Josua Krause, Adam Perer, and Enrico Bertini. 2014. INFUSE : Interactive Feature Selection for Predictive Modeling of High Dimensional Data. *IEEE Transactions on Visualization and Computer Graphics* 20, 1 (2014), 1614–1623. <https://doi.org/10.1109/TVCG.2014.2346482>
- [16] B. Jaya Lakshmi, M. Shashi, and K. B. Madhuri. 2020. A rough set based subspace clustering technique for high dimensional data. *Journal of King Saud University - Computer and Information Sciences* 32, 3 (2020), 329–334. <https://doi.org/10.1016/j.jksuci.2017.09.003>
- [17] Jihye Lee, Hyouji Ha, Hyunwoo Han, Sungyun Bae, Sangjoon Son, and Kyungwon Lee. 2017. Visual Analytics for Biomedical Cluster Subdivision : A Design Study with Psychiatrists. In *Computer Graphics International Conference*. 1–6. <https://doi.org/10.1145/3095140.3095157>
- [18] Stuart P. Lloyd. 1982. Least Squares Quantization in PCM. *IEEE Transactions on Information Theory* 28, 2 (1982), 129–137. <https://doi.org/10.1109/TVT.1982.1056489>
- [19] Javier Luraschi and JJ Allaire. 2018. r2d3: Interface to 'D3' Visualizations. <https://cran.r-project.org/package=r2d3>
- [20] Lance Parsons, Ehtesham Haque, and Huan Liu. 2016. Subspace Clustering on High Dimensional Data. *i-manager's Journal on Cloud Computing* 3, 3 (2016), 1–18. <https://doi.org/10.26634/jcc.3.3.8297>
- [21] Peter J. Rousseeuw. 1987. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *J. Comput. Appl. Math.* 20, C (1987), 53–65. [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
- [22] Carson Sievert. 2020. Interactive Web-Based Data Visualization with R, plotly, and shiny. <https://plotly-r.com>
- [23] Michael Steinbach, Levent Ertöz, and Vipin Kumar. 2003. The Challenges of Clustering High Dimensional Data. In *New Directions in Statistical Physics*. 273–309. <https://doi.org/10.1007/978-3-662-08968-2>
- [24] R Development Core Team. 2010. R: A Language and Environment for Statistical Computing. <http://www.r-project.org/>
- [25] Michael Thrun, Felix Pape, and Alfred Ultsch. 2020. Interactive Machine Learning Tool for Clustering in Visual Analytics. In *IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA)*. 479–487. <https://doi.org/10.1109/DSAA49011.2020.00062>
- [26] Robert Tibshirani, Guenther Walther, and Trevor Hastie. 2001. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society. Series B: Statistical Methodology* 63, 2 (2001), 411–423. <https://doi.org/10.1111/1467-9868.00293>